①

$\mathbb{P}$: physical world → forecast → real data

$\mathbb{Q}$: risk neutral → pricing → consumer motivation

PCA ⟶ X_t fattori latenti di rischio

$N = 3 \begin{cases} X_1 = \text{livello} \\ X_2 = \text{pendenza} \\ X_3 = \text{curvatura} \end{cases}$ Var spiegata $> 95\%$

zero coupon yield

1) $\Delta X_t = K_0^{\mathbb{Q}} + K_1^{\mathbb{Q}} X_{t-1} + \Sigma \epsilon_t^{\mathbb{Q}}$

2) $X_t = K_0 + (I + K_1) X_{t-1} + \Sigma \epsilon_t$

- $\epsilon_t \sim N(0, I_N)$
- Σ : sigma varianza (fac di triangolare inferiore, Cholesky)

3) $s_t = \rho + \rho_1' X_t$

shadow rate

4) $r_t = \max\{s_t, LB\}$

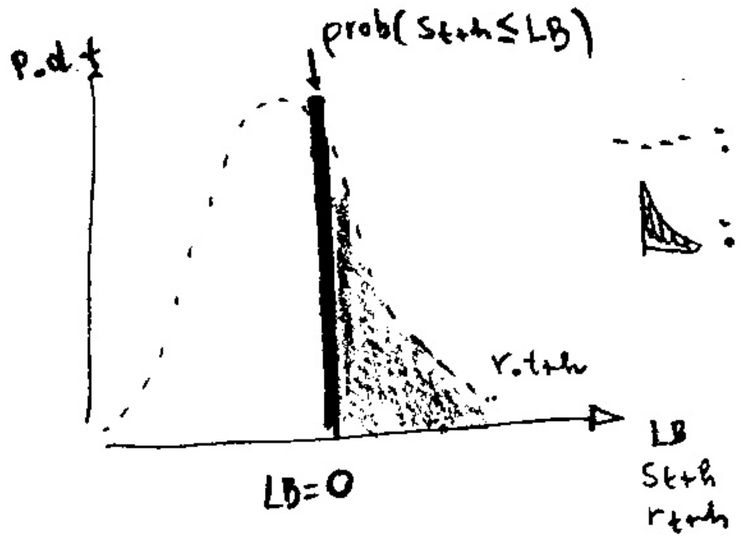
observable rate

SRISM

$\Rightarrow r_t$ can not reach not plausible negative values: $r_t = \rho_0 + \rho_1' X_t$

Paper: EONIA OS based $\geq$ DF rate of ECB

4) $\Rightarrow$



ex: LB=0

Prob is censored

vs.

ATSM $y_t = A + B \tilde{X}_t$ (rendimenti; $\tilde{X}_t$: rischio)

PROBLEM: can go so much negative

$\Rightarrow P(r_{t+h} = LB) = \mathbb{P}(s_{t+h} \leq LB) = \Phi\left(\frac{LB - \mu_{t,h}}{\sigma_h}\right)$

Φ: C.D.F cumulata

NOTATION: $\begin{cases} \mu_{t,h} = \mathbb{E}_t(s_{t+h}) & \text{time, hor dependent} \\ (\sigma_h)^2 = Var(s_{t+h}) & \text{time invariant} \end{cases}$

$\Rightarrow$ 5) $\mathbb{E}_t(r_{t+h}) = LB + \sigma_h \cdot H\left(\frac{\mu_{t,h} - LB}{\sigma_h}\right)$

Standard result

NOTE: $\begin{cases} H(x) = x \cdot \Phi(x) + \phi(x) \\ \Phi : \text{CDF} \\ \phi : \text{PDF} \end{cases}$

PRICE at t of ZeroCoupon that pay 1 at t+1 := P_t^n

(value attualizzato continuo) $P_t^n = \mathbb{E}_t^{\mathbb{Q}}\left(e^{-\sum_{i=0}^{n-1} r_{t+i}}\right)$ 6)

$\Rightarrow$ Bond Yields for maturity n: $y_t^n = \frac{1}{n} \ln P_t^n$

50 : find a g_n :

$$y_t^n = g_n(X_t; K_0^Q, K_1^Q, \Sigma, \rho_0, \rho_1, LB) \qquad 7)$$

SRTSM

$$y_t^n = \frac{1}{n} \ln P_n^t = \frac{1}{n} \ln \left(\mathbb{E}_t^Q \left(e^{-\sum_{i=0}^{n-1} r_{t+i}} \right) \right)$$

with 4) $$= \frac{1}{n} \ln \left(\mathbb{E}_t^Q \left(e^{-\sum_{i=0}^{n-1} \max\{s_{t+i}, LB\}} \right) \right)$$

$$= \frac{1}{n} \ln \left(\mathbb{E}_t^Q \left(e^{-\sum_{i=0}^{n-1} \max\{\rho_0 + \rho_1' X_{t+i}, LB\}} \right) \right)$$

$$= \frac{1}{n} \ln \left(\mathbb{E}_t^Q \left(e^{-\sum_{i=0}^{n-1} \max\{\rho_0 + \rho_1' (\underbrace{K_0 + (I + K_1) X_{t+i-1} + \Sigma \epsilon_{t+i}}_{X_t}), LB\}} \right) \right)$$

Parameters :

Recall (NO 4)
$r_t = \rho_0 + \rho_1' X_t$
8

without 4) $$= \cdots \cdots \cdots \cdots e^{-\sum_{i=0}^{n-1} r_{t+i}}$$

$$= \cdots \quad e^{-\sum_{i=0}^{n-1} \rho_0 + \rho_1' X_{t+i}}$$

ATSM

$$= \cdots \Rightarrow \quad y_t^n = A_n + B_n' X_t \qquad 9)$$

where : $\Delta B_n = K_1^{Q'} B_{n-1} - \rho_1$ with $A_0 = 0$

$\Delta A_n = K_0^{Q'} B_{n-1} + \frac{1}{2} B_{n-1}' \Sigma \Sigma' B_{n-1} - \rho_0$ $\quad B_0 = 0$

IF $N > 1 \Rightarrow$ NO Analytical solution !

è una correzione disuguaglianza Jensen variance term

WU XIA : approx

$$y_t^n \approx \tilde{g}_n(\,) = \frac{1}{n} \left(r_t + (n-1) LB + \sum_{h=1}^{n-1} \sigma_h^Q \cdot H\left(\frac{\mu_{t,h}^Q - LB - J_h}{\sigma_h^Q} \right) \right) \qquad 13)$$

$$\mu_{t,h}^Q = \mathbb{E}_t^Q(s_{t+h}) =_{\text{as } 5)} LB + \sigma_h \cdot H\left(\frac{\mu_{t,h} - LB}{\sigma_h} \right)$$

$$\sigma_h^Q = Var_t^Q(s_{t+h})$$

JSZ

for SRTSM : $K_1^Q = \begin{pmatrix} K_{1,11}^Q & 0 & 0 \\ 0 & K_{1,22}^Q & 0 \\ 0 & 0 & K_{1,33}^Q \end{pmatrix}$ and $K_0^Q = \begin{pmatrix} K_{0,1}^Q \\ 0 \\ 0 \end{pmatrix}$ and Σ lower triang.

autoreg matrix

assumed that $\rho_0 = 0$ and $\rho_1 = (1, 1, 1)$

for $\mathbb{P}$: K_0^P, K_1^P free.

ESTIMATION

- Possible shift in LB during period:

$$4 \Rightarrow \qquad r_t = \max\{s_t, LB_t\} \qquad (14)$$

estimating SRTSM: state space form:

TRANSITION Eq: linear gaussian factor Dynamics

$$X_t = K_0^P + (I_N + K_1^P) X_{t-1} + \Sigma \epsilon_t^P \quad , \quad \epsilon_t^P \overset{iid}{\sim} N(0, I_N) \qquad (15)$$

$\Rightarrow$

$$16) \qquad \begin{pmatrix} y_t^{3m,0} \\ \vdots \\ y_t^{10y,0} \end{pmatrix} = \begin{pmatrix} \tilde{g}_{3m}(X_t; K_0^Q, K_1^Q, \Sigma, \rho_0, \rho_1, LB_t) \\ \vdots \\ \tilde{g}_{10y}(\cdot \quad \cdot\cdot) \end{pmatrix} + \begin{pmatrix} \epsilon_t^{3m} \\ \vdots \\ \epsilon_t^{10y} \end{pmatrix} \qquad \epsilon_t^n \overset{iid}{\sim} N(0, \sigma_e^2)$$

$\tilde{g}_n$ as in 13)

Parameters: $\psi = (K_{0,1}^Q, K_{1,11}^Q, K_{1,22}^Q, K_{1,33}^Q, K_{0,1}^P, K_{1,11}^P, K_{1,22}^P, K_{1,33}^P, \Sigma_{11}, \Sigma_{21}, \ldots, \Sigma_{33}, \ldots \underbrace{LB_A, LB_B, LB_C}, \sigma_e)$ (17)

A: 1999 → April 14
B: May 14 → Aug 14
C: Sep 14 → Jun 15

26 elements with 3 MLB periods abstract

SSRTSM and ASM

new latent factors: $\mathcal{P}_t \Rightarrow$ affine: $\mathcal{P}_t = \rho_0 + \rho_1 X_t$ ← PCA

→ apply filtered SRTSM latent factors X_t → extract SRTSM factor $\mathcal{P}_t$:

$J = 8$ curves yields mapped in 3 obs PCA's: $P_t^0 = W y_t^0$ (18)

notice loading

$$y_t^0 = \begin{pmatrix} y_t^{0,3m} \\ \vdots \\ y_t^{0,10y} \end{pmatrix}$$

Initial queries: • $\mathbb{Q}$ dynamics of X_t from 1)

• loading A_n, B_n from 9)

$$\Downarrow$$

ATSM: $Y_t = A + BX_t \Rightarrow P_t = W y_t = W(A + BX_t) = WA + WBX_t$ (19)

new latent factors: affine transformation

Remember: Affine $\equiv$ linear.

20) $\Rightarrow y_t = A_P + B'_P P_t$ $\begin{cases} A_P = (I - B(WB)^{-1}W)A \\ B_P = B(WB)^{-1} \end{cases}$

SRTSM: identified of factors dynamics as described above:

$$P_t = WA + WBX_t \qquad (21)$$

W, as 18) — prese da ATSM

A, B as 19) — taken from ATSM estimation

see: 17) Ψ: P_t has properties of 1), 2):

$\mathbb{Q}$ $\begin{cases} K_{0,P} = WBK_0 - WBK_1(WB)^{-1}WA \\ K_{1,P} = WBK_1(WB)^{-1} \\ \Sigma_P \Sigma'_P = WB\Sigma\Sigma'(WB)' \end{cases}$

$\begin{cases} \rho_{0,P} = \rho_0 - \rho'_1(WB)^{-1}WA \\ \rho_{1,P} = ((WB)^{-1})'\rho_1 \end{cases}$

(if this is done for ATSM $\rho_{0,P}$ e $\rho_{1,P}$ map into short rate and not shadow)

$\begin{pmatrix} JSZ \\ \mathbb{Q} \end{pmatrix} \rightarrow \Delta X_t = \dots \rightarrow K_1^{\mathbb{Q}}$ dual

δ_t.

Kalman extended

EFK non linear: find Jacobian for ΔX_t and $\tilde{y}^n() = \dots(H()) \dots$